

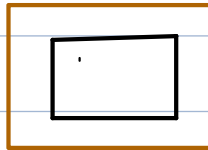
A — Wifi Router — ISP Network — R1 — R2 — Dest Network — B

Application

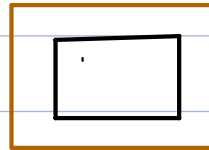
{ Application
Presentation
Session



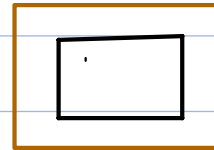
Transport



Segment



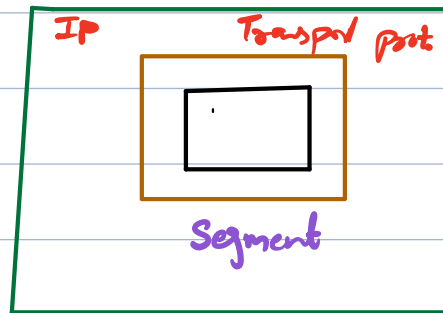
Segment



Segment

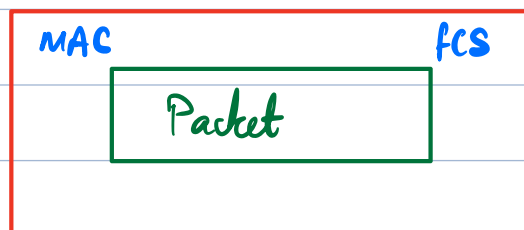


Network

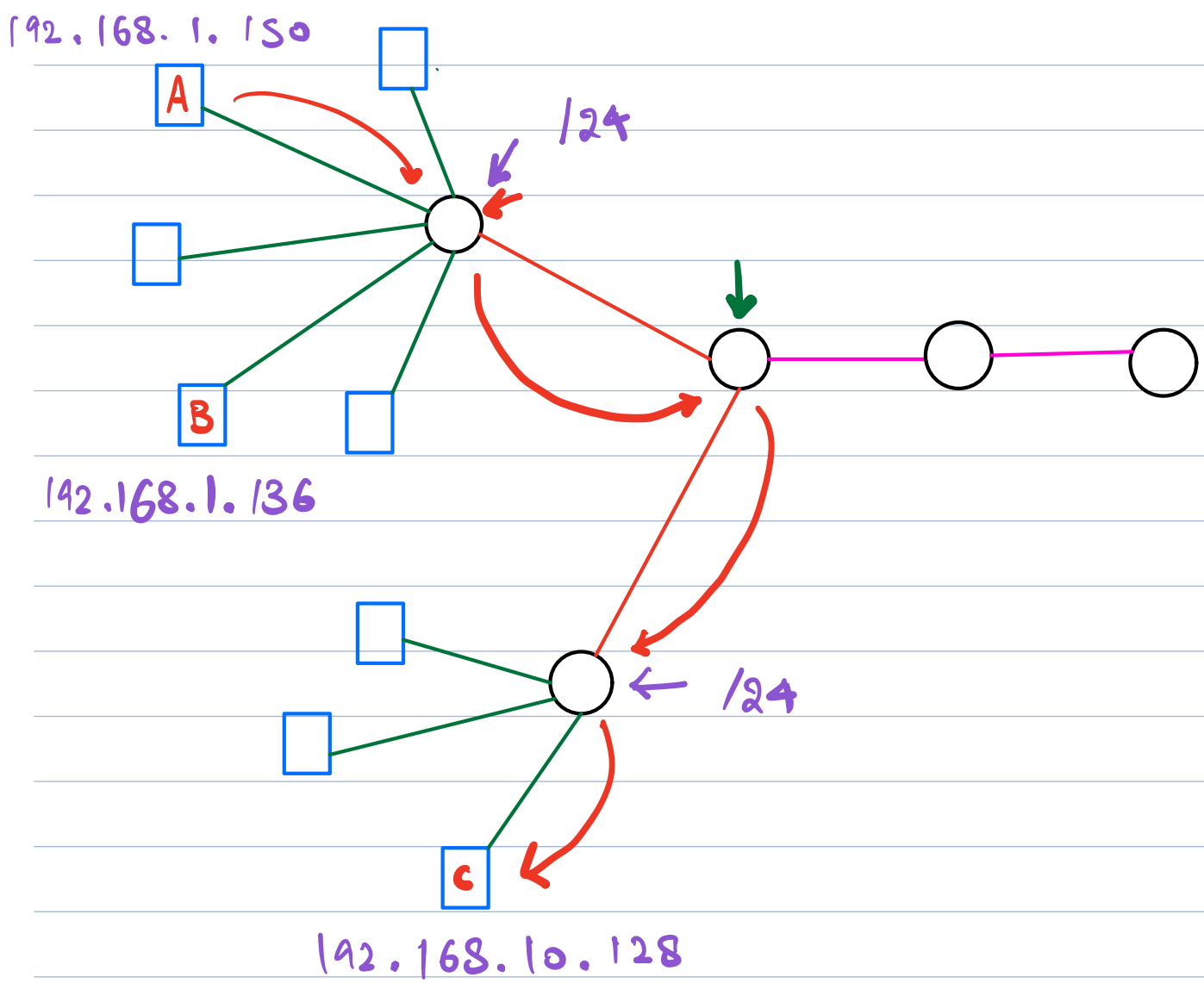


Packet

Data link



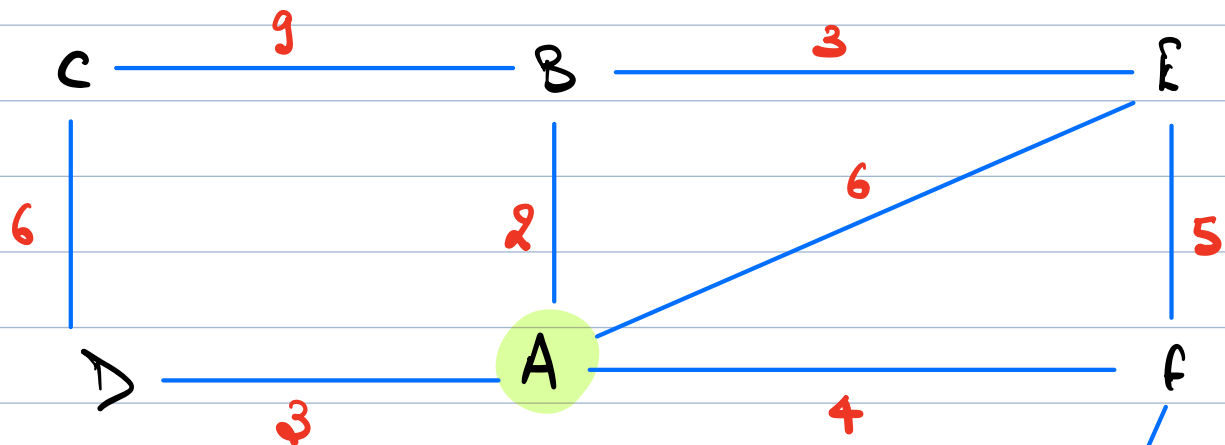
frame



- 1) Dijkstra's Algo
- 2) Bellman ford Algo
- 3) MST & their network applications

MinHep { extractMin() } $O(\log N)$
 { insert() }

Dijkstra's Algo



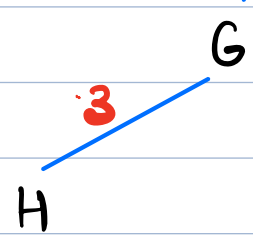
A	B	C	D	E	F	G	H
0	2	9	3	5	4	10	13

Minheap $\langle \text{dist}, \text{Node} \rangle$

- ~~$\langle 0, A \rangle$~~
- ~~$\langle 3, D \rangle$~~
- ~~$\langle 2, B \rangle$~~
- ~~$\langle 6, E \rangle$~~
- ~~$\langle 4, F \rangle$~~
- ~~$\langle 11, C \rangle$~~
- ~~$\langle 5, E \rangle$~~
- ~~$\langle 9, C \rangle$~~
- ~~$\langle 10, G \rangle$~~
- ~~$\langle 13, H \rangle$~~

~~$\langle 11, C \rangle$~~

$\langle 13, H \rangle$
 $\hookrightarrow G = 13 + 3 = 16$ ✗



$\langle 0, A \rangle$

- $\rightarrow D = 3$
- $\rightarrow B = 2$
- $\rightarrow E = 6$
- $\rightarrow F = 4$

$\langle 2, B \rangle$

- $\rightarrow C = 2 + 9 = 11$
- $\rightarrow E = 2 + 3 = 5$
- $\rightarrow A = 2 + 2 = 4$ ✗

$\langle 3, D \rangle$

- $\rightarrow C = 3 + 6 = 9$
- $\rightarrow A = 3 + 3 = 6$ ✗

$\langle 4, f \rangle$

$$\begin{aligned} \rightarrow E &= 4 + 5 = 9 \quad \times \\ \rightarrow A &= 4 + 4 = 8 \quad \times \\ \rightarrow G &= 4 + 6 = 10 \end{aligned}$$

$\langle 5, E \rangle$

$$\begin{aligned} \rightarrow A &= 5 + 6 = 11 \quad \times \\ \rightarrow B &= 5 + 3 = 8 \quad \times \\ \rightarrow F &= 5 + 5 = 10 \quad \times \end{aligned}$$

$\langle \underline{6}, E \rangle \quad d_E > \text{Shortest}(E)$

$\langle 9, C \rangle$

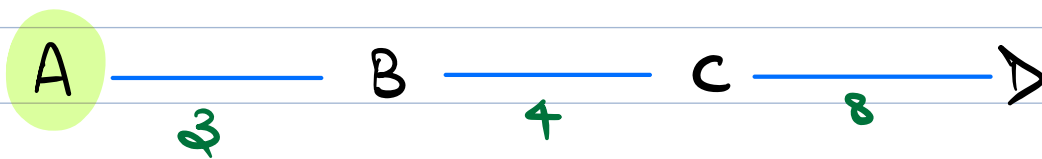
$$\begin{aligned} \rightarrow D &= \quad \times \\ \rightarrow B &= \quad \times \end{aligned}$$

$\langle 10, G \rangle$

$$\begin{aligned} \rightarrow F &= 10 + 6 = 16 \quad \times \\ \rightarrow H &= 10 + 3 = 13 \end{aligned}$$

Belmann ford Algo

Relax all edges $N-1$ times



A	B	C	D
0	3	7	15

✓ C-D, 8

✓ B-C, 4

✓ A-B, 3

C-D

Relax $A \rightarrow D$ via C-D

$\text{dist}(C) + W_{C-D}$

$$7 \cancel{\neq} + 8 = \cancel{15}$$

B-C

Relax $A \rightarrow C$ via B-C

$\text{dist}(B) + W_{B-C}$

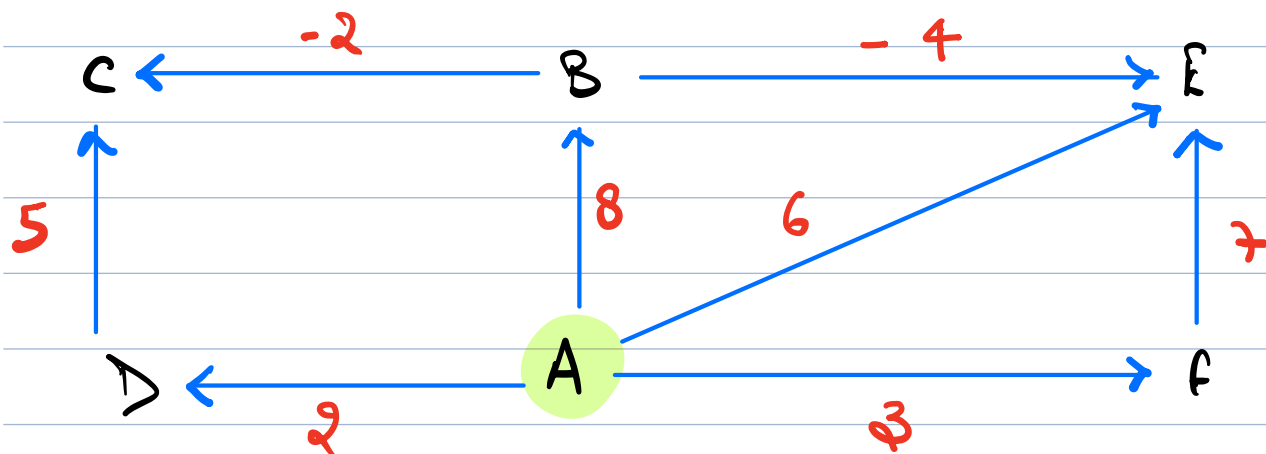
$$3 \cancel{\neq} + 4 = \cancel{7}$$

A-B

Relax $A \rightarrow B$ via A-B

$\text{dist}(A) + W_{A-B}$

$$0 + 3 =$$



A	B	C	D	E	F
0	8	6	2	4	3

$$\times \quad A - B, 8$$

$$0 + 8 = 8$$

$$\times \quad A - E, 6$$

$$0 + 6 = 6$$

$$\times \quad A - D, 2$$

$$0 + 2 = 2$$

$$\times \quad B - C, -2$$

$$8 + (-2) = 6$$

$$\times \quad B - E, -4$$

$$8 + (-4) = 4$$

$$\times \quad F - E, 7$$

$$3 + 7 = 10$$

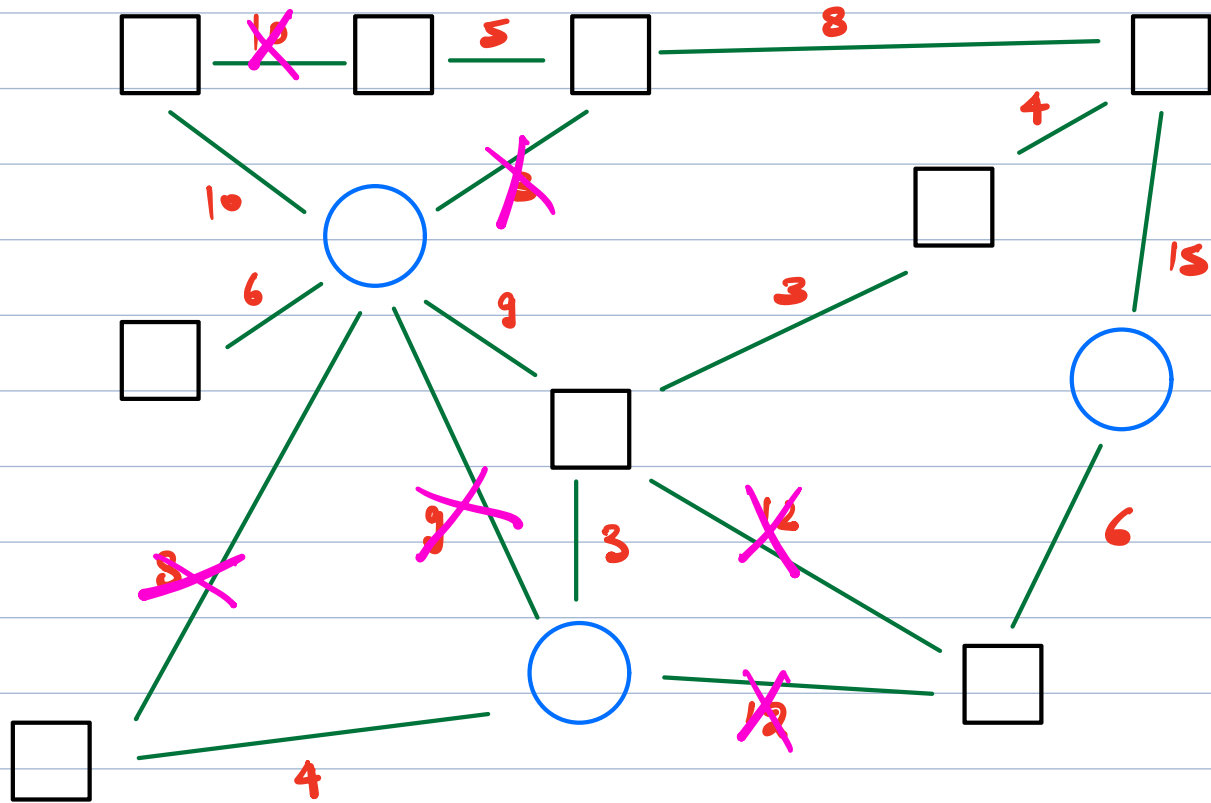
$$\times \quad A - F, 3$$

$$0 + 3 = 3$$

$$\times \quad D - C, 5$$

$$2 + 5 = 7$$

MST $\Rightarrow$ Minimum Spanning tree



N nodes $\Rightarrow (N-1)$ Edges.